Secure Exit

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT3-Debugging and Optimization

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QUESTIONS

Q1
ATM Withdrawal – Exception Handling Debugging
10 marks

Q2
Hospital Registration – NullPointerException Debugging
10 marks

Q3
E-Commerce Product Search – Array Exception Debugging
10 marks

Q4
Bank Account – Race Condition Debugging
10 marks

Q5
Railway Reservation – Synchronization Debugging
10 marks

5/5 answered

Q1 **ATM Withdrawal – Exception Handling Debugging** 10 marks

An ATM application is used to process cash withdrawals. The following Java program contains programming errors.

Task:

1. Run the given program and identify the errors.
2. Debug and modify the program.
3. Submit the COMPLETE corrected Java program.
4. The corrected program must handle insufficient balance, zero withdrawal, invalid input, and unexpected errors.
5. Use appropriate try-catch-finally exception handling.
6. The corrected program must execute successfully for all valid test cases.

Faulty Program:

```
import java.util.Scanner;

class ATM {
public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

System.out.print("Enter account balance: ");
int balance = sc.nextInt();
```

```
System.out.print("Enter withdrawal amount: ");
int amount = sc.nextInt();

try {
int remaining = balance / amount;

if (amount > balance) {
System.out.println("Insufficient balance");
} else {
System.out.println("Withdrawal successful");
System.out.println("Remaining balance = "
+ (balance - amount));
}
}
catch (ArithmeticException e) {
System.out.println("Transaction failed");
}

System.out.println("Transaction completed");
}
}
```

Expected correction:
The program must correctly validate the withdrawal amount before performing the transaction, handle division-by-zero and invalid input, and always display "Transaction completed" after exception handling.

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
import java.util.Scanner;

class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        try{
            int balance = sc.nextInt();
```

```
            int amount  = sc.nextInt();

            if(amount<=0){
                throw new IllegalArgumentExc
            }
            if(amount > balance){
                System.out.println("Insuffic
            }else{
                System.out.println("Withdraw
                System.out.println("Remainin
            }
        }catch(ArithmeticException e){
            System.out.println("Transaction
        }catch(java.util.InputMismatchExcept
            System.out.println("Invalid inpu
        }catch(IllegalArgumentException e){
            System.out.println("Invalid with
        }catch(Exception e){
            System.out.println("Unexpected e
        }finally{
            System.out.println("Transaction
            sc.close();
        }
    }
}
```

Test Input

```
abc
```

Output

```
Invalid input. Please enter integers only.
Transaction completed
```

Pending manual review

Submitted